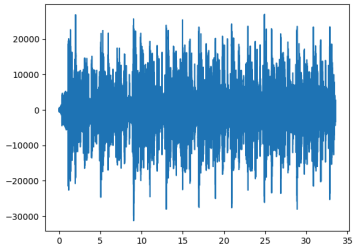
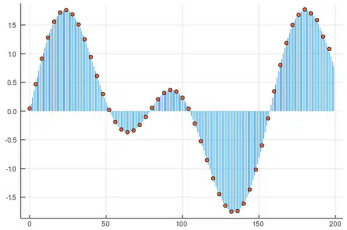
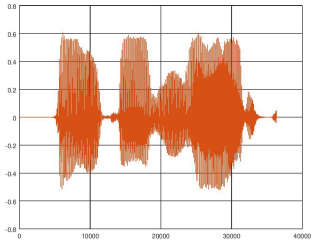


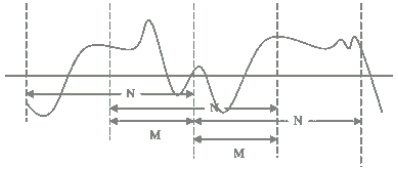
Waveform



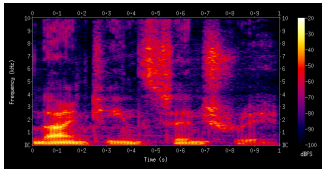
Resampling / Channel Unification



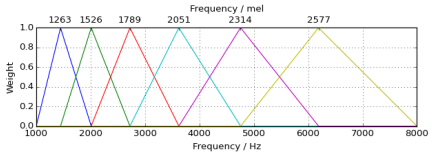
Amplitude Normalization



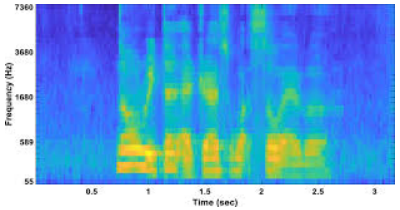
Framing & Windowing



STFT



Mel Filterbank



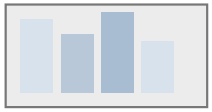
log-Mel

CMVN / Standardization

Padding / Truncation

Noise / Shift / Stretch

Normalization / Length Handling / Augmentation



Model Input